# Ojasva Mishra

Z Fellow | Software Engineering | AI/ML | Quantitative Research | Robotics
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## Education

**University of Florida** — Expected May 2029
*B.S. Computer Engineering and Finance | GPA: 4.0/4.0 | YC Startup School 2026 | SAT 1590, National Merit* — *Gainesville, FL*

## Experience

**Carnegie Mellon University, Xu Lab** — Jul 2025 – Present
*Research Intern* — *Pittsburgh, PA*
- Tuned PID gains: **49% less tracking error**, under 2% overshoot at torque limits; conformal filters reject spoofed sensor data.
- Validated the controllers on a Franka Emika Panda arm with Prof. Min Xu and Dr. Xiaolong Wu; **first author on 3 papers**.

**Drift (Antler-backed)** — Nov 2025 – Aug 2026
*AI/Software Engineer (Founding Team) | TypeScript, Rust, ROS2* — *San Francisco, CA*
- Built the **Gaussian splat pipeline** that turns photos of real spaces into 3D robot sim worlds (ROS2, Gazebo, Isaac Sim).
- Built a self-healing loop that triages user-reported errors and auto-patches code: auto-fix rate **12% to 60%** over 3,000+ errors.
- Built core infra and internal tools (TypeScript, Rust, Tauri, React) for 1,000+ users, plus LLM evals graded by a vision model.
- Traced a production Gazebo freeze to gz-transport UDP multicast failing on Ubuntu 24.04, then shipped the fix.

**Twinly** — Dec 2025 – Present
*Co-Founder and CEO (Z Fellows cohort) | Swift, TypeScript, Python, Supabase* — *San Francisco, CA*
- Modeled each user's writing from 4,000 sent messages (SQLite, IMAP) and cloned their voice on-device (MLX).
- Built the Swift agent that runs each clone's tasks on a real computer via the macOS accessibility tree, a CDP-driven headless Chrome, and screen vision; shipped on macOS, Windows, and iOS to **1,400+ active users, including paying customers**.

**AlgoGators Investment Fund** — Sep 2026 – Present
*Quantitative Research Analyst* — *Gainesville, FL*
- 1 of 2 first-years on the quant team of a ~$500K AUM fund; ran a crack-spread strategy: **64% win rate over 1,200 live trades**.

**EV Bots** — Apr 2025 – Feb 2026
*AI Robotics Engineer Intern* — *Chester Springs, PA*
- Built the first autonomous EV-charging robot (provisional patents): embedded control code (Arduino), electronics, sensors.
- Closed the perception-to-actuation loop: **100% plug-in success over 30 trials**, now deployed; also built a drone dock.

**PteroLabs** — Jul 2024 – Jul 2025
*Software Engineering Intern* — *Remote*
- Built a UE5 mission drone simulator with PX4, ArduPilot, and Betaflight autopilots in the loop, adopted at UPenn.

**iPipeline** — Jun 2025 – Aug 2025
*Software Development Intern | TypeScript, Python, Docker* — *Exton, PA*
- Shipped automated test frameworks, AI agents, and developer tooling for iGO, the life insurance e-application (Python, Docker).

## Research and Publications

- Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. *T-RO* (accepted)
- Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. *ICRA 2027* (accepted)
- SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control. *IROS 2026* (under review)

## Projects and Competitions

**Déjà View** | 1st in 2 tracks, SwampHacks XII | *TypeScript, three.js, WebXR, Python, COLMAP, gsplat, Luau* | Live | GitHub
- Trained a 1M-Gaussian splat from COLMAP poses in 11 min and exported the scene to Meta Quest, web, and Roblox.

**Courtside** | Gator Quant Hacks 2026 | *Python, OpenCV, NumPy, pandas, SciPy, DuckDB, WebSockets* | Paper | GitHub
- Tested a pre-registered latency-arbitrage hypothesis on 13,084 Polymarket tennis markets: wallets trading within 3 s of a point were **profitable after fees in all 11 out-of-sample months** (Sharpe 6.7 over 1,500+ trades; deflated Sharpe, block bootstrap).
- Built a causal computer-vision ball tracker that calls points a median **162 ms before the bounce** at 119.9 fps, none wrong.

**Lucid Dreams** | 1st place, Gator Hack V | *TypeScript, three.js, WebXR, PyTorch, gsplat, Unity, Stockfish* | Live | GitHub
- Turned a phone video into a life-size 3D person in Quest: 48 generated views, ~290K Gaussians a frame on **12 NVIDIA B200s**.
- Made avatars interactive: SMPL-X rigs with two-bone IK, a WASM Stockfish opponent, 240 Hz ball physics, 3.1 s voice replies.

**More projects and awards**
- **Dream Team Engineering, NeuroSpine:** with UF Neurosurgery, building nnU-Net spine CT/MRI segmentation for bone quality.
- **Volition** (EEG exoskeleton; Python, CadQuery): designed the full CAD and a 1 kHz torque-limited control sim that walks 8 steps.
- **VEX Robotics, 3x Worlds** (captain, lead programmer); **TSA, 2x National Finalist** (engineering design, drones, VR).

## Technical Skills

**Languages:** Python, C++, C, TypeScript, JavaScript, Swift, Rust, Java, C#, SQL, MATLAB, Bash, Luau
**Frameworks and libraries:** React, Next.js, Node.js, FastAPI, PyTorch, Hugging Face, scikit-learn, OpenCV, three.js, SwiftUI
**Developer tools and cloud:** Git, Docker, AWS, Azure, GCP, PostgreSQL, Supabase, CUDA, Slurm, Claude Code, Codex, Cursor
**Quant:** Zetamac 142, statistics (STA3032), backtesting, Markov models, deflated Sharpe, block bootstrap, NumPy, pandas
**Robotics and hardware:** ROS2, Gazebo, MuJoCo, Isaac Sim, PX4, SLAM, PID, Arduino, RTOS, PCB design, I2C/SPI/UART, CAD